Representation Learning from MSI using VAEs



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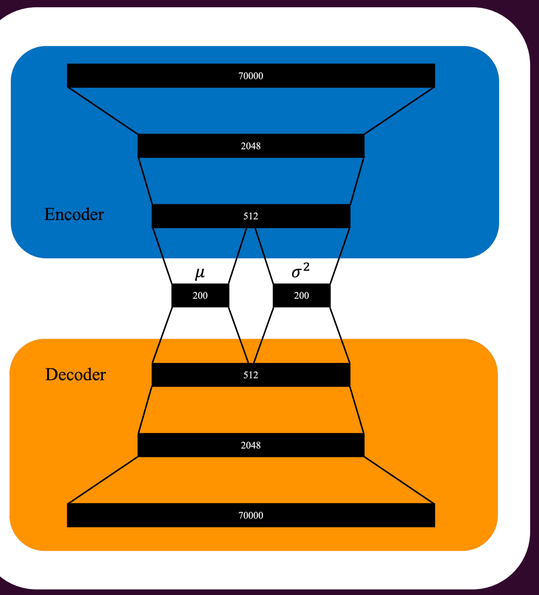
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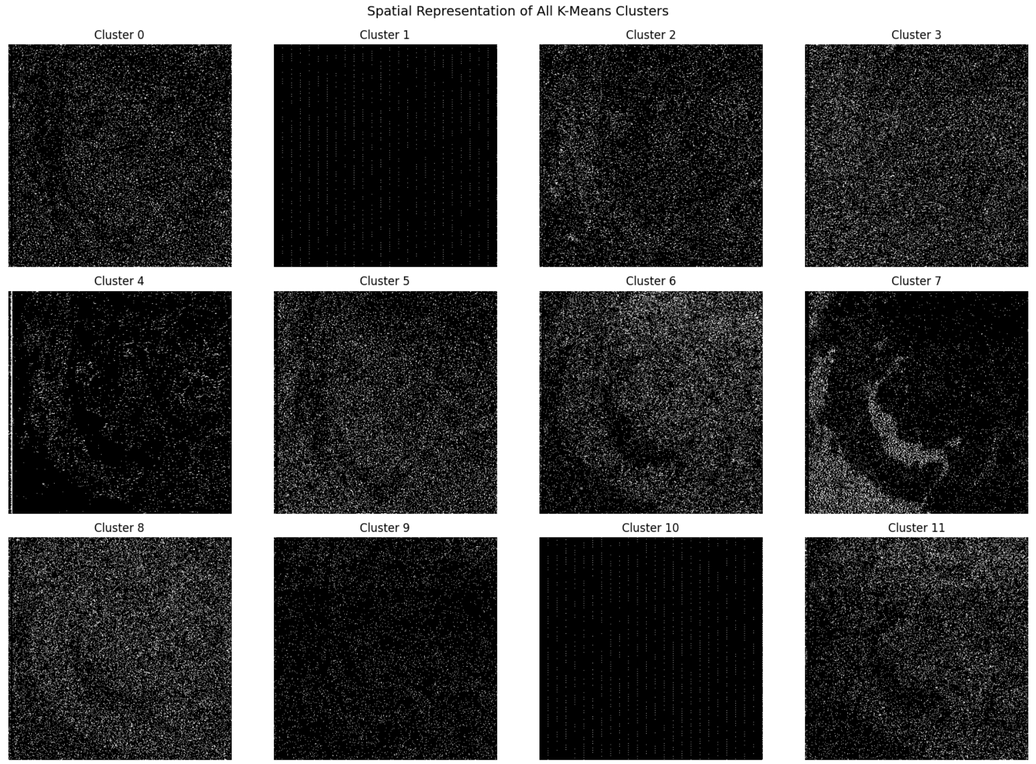
Background

Mass spectrometry imaging (MSI) has seen many improvements in recent years, one of which is an increase in spatial resolution which has led to the need for better tools for data visualisation and data analysis [1]. The use of autoencoders as a method to compress MSI data was first introduced by Thomas et al. [2]. This only allowed for learning a discrete non-regularised latent space, and therefore in order to make use of a model with generative properties, a variational autoencoder (VAE) architecture was employed.

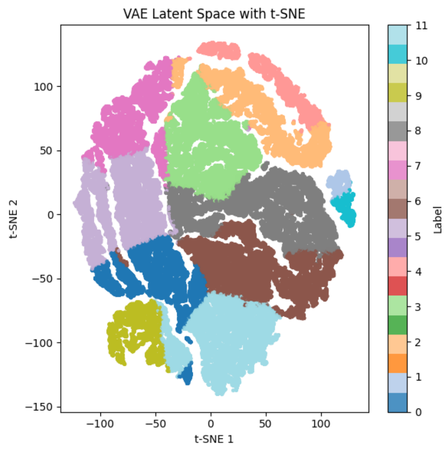
Method

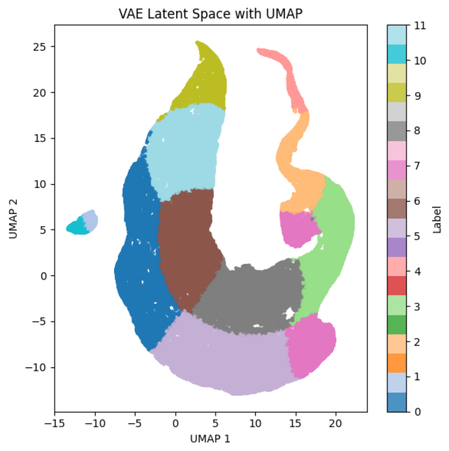
The MSI data was passed through different variations of a variational autoencoder. The latent spaces that were learnt were clustered using K-means clustering. The latent spaces were visualised using UMAP and t-SNE. The clusters were overalayed on these plots. The spatial distribution of the clusters were also evaluated in order to link any learnt representations to observations in the tissue that was imaged.





Out of the 12 clusters learnt, few showed potential at representing meaningul information in the spatial domain. Cluster 7 highlights that some of the latent space was able to capture meaningful information.

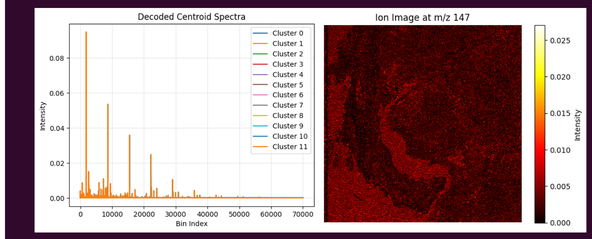




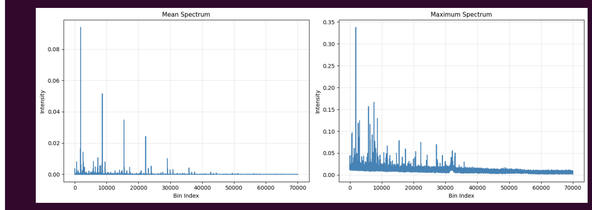
Data Description

The training data for the VAE was made up of 160 000 spectra that were captured in a 400 by 400 grid. Each spectra consisted of intensity readings at different mass-to-charge values. The intensity readings were binned uniformly into 70 000 bins. This is why the VAE input took in 70 000 values.

The cluster centroids obtained by running K-Means on the latent space were passed through the decoder of the VAE and the produced spectra are shown below. The spectra look extremely similar but when the MSE was measured between the output, differences were noted. The figure on the right shows the spatial representation of different intensity values across spectra at a specific mass-to-charge value.



Due to the latent space representing very similar spectra, the average and maximum values for the different bin indices were recorded across all of the spectra and are plotted below. It is important to note that the mean spectrum is very similar to the spectra that were being represented in the latent space.



Key References

[1] Y. Zhang and X. Liu, “Machine learning techniques for mass spectrometry imaging data analysis and applica-tions,” pp. 519–522, 2018.
[2] S. A. Thomas, A. M. Race, R. T. Steven, I. S. Gilmore,and J. Bunch, “Dimensionality reduction of mass spec-trometry imaging data using autoencoders,” in 2016 IEEE symposium series on computational intelligence (SSCI). IEEE, 2016, pp. 1–7.

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